Abstract

on

Full Stack Online Learning Platform

An abstract submitted for

Project Phase-I

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ABSTRACT

The rapid growth of digital education has created a need for flexible, accessible, and scalable online learning platforms that allow students and instructors to interact through a unified digital environment. This project proposes the design and development of a **Full-Stack Online Learning Platform Using Spring Boot, Docker and Kubernetes**. The platform is designed to provide students with an efficient environment for discovering courses, enrolling in courses, accessing learning materials, completing quizzes, and tracking their learning progress.

The backend of the system is developed using **Spring Boot**, which provides **RESTful APIs** for user authentication, course management, enrollment, lesson management, quizzes, and progress tracking. The platform supports different user roles, including students, instructors, and administrators, with role-based access to their respective functionalities. A web-based frontend provides an interactive interface through which users can access the platform's services, while a relational database is used to securely store user, course, enrollment, assessment, and progress-related information.

To improve portability and deployment efficiency, the application components are packaged into **Docker containers**. **Kubernetes** is used for container orchestration, enabling efficient deployment, service management, scaling, and improved availability of the application. The system follows a modular architecture that allows its components to be independently managed and deployed.

The proposed platform demonstrates the integration of modern full-stack development with containerization and orchestration technologies to create a scalable and maintainable online learning environment. The project provides practical implementation of **Spring Boot, RESTful APIs, database management, authentication, Docker, Kubernetes, and cloud-oriented deployment**, making the system suitable for modern educational applications.

Keywords: Online Learning, Full-Stack Development, Spring Boot, REST API, Docker, Kubernetes, E-Learning, Containerization, Cloud Deployment.

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